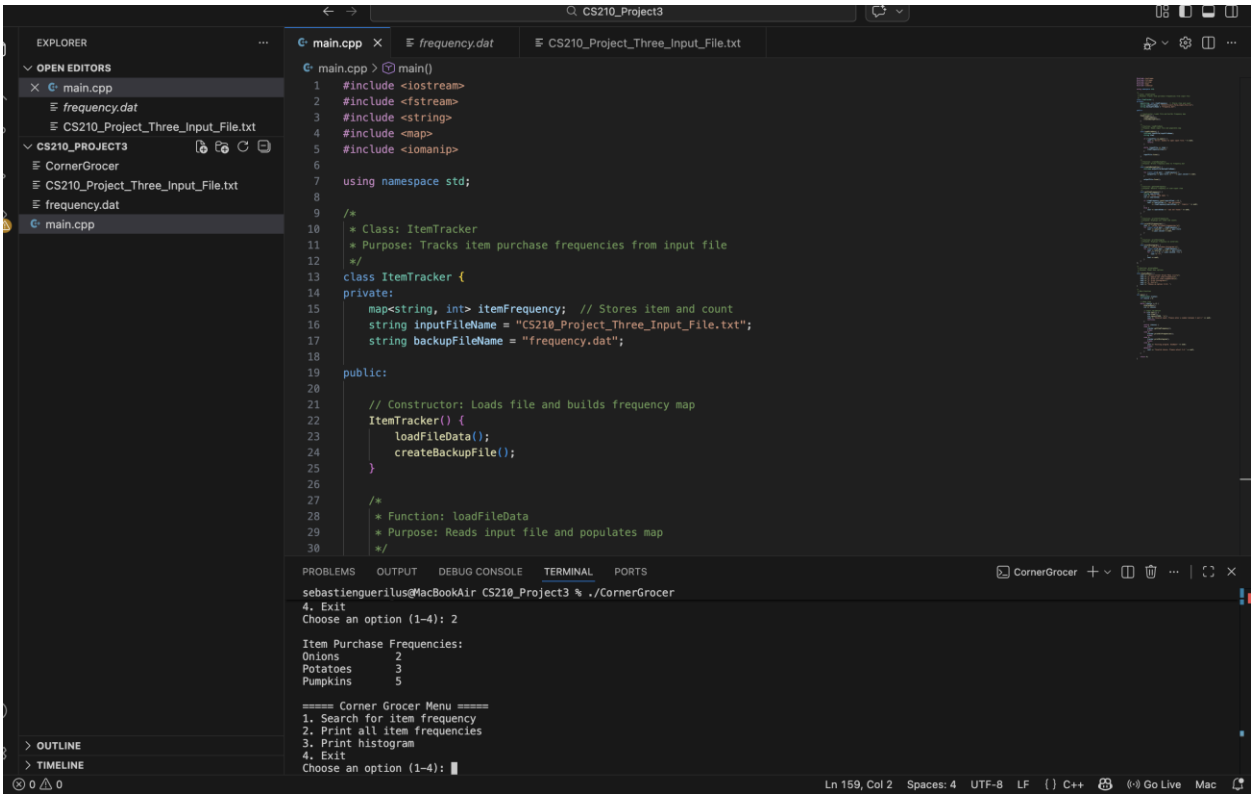
Project Summary

This program reads grocery item data from a file and calculates how many times each item was purchased. The data is stored using C++ map for efficient counting and retrieval.

The program features a menu system allowing users to: Search for a specific item frequency. Display all item frequencies. Display a histogram representation. Exit the program. A backup file (frequency.dat) is automatically generated at program start.

- A class was used to ensure modularity and encapsulation.
- A map was selected because it allows automatic sorting and efficient lookup.
- Input validation was implemented to prevent runtime errors.
- The program uses clear naming conventions and comments for maintainability.



EXPLORER

OPEN EDITORS

- main.cpp
- frequency.dat
- CS210_Project_Three_Input_File.txt

CS210_PROJECT3

- CornerGrocer
- CS210_Project_Three_Input_File.txt
- frequency.dat
- main.cpp

main.cpp > main()

```cpp
#include <iostream>
#include <fstream>
#include <string>
#include <map>
#include <iomanip>

using namespace std;

/*
 * Class: ItemTracker
 * Purpose: Tracks item purchase frequencies from input file
 */
class ItemTracker {
private:
    map<string, int> itemFrequency;  // Stores item and count
    string inputFileName = "CS210_Project_Three_Input_File.txt";
    string backupFileName = "frequency.dat";

public:

    // Constructor: Loads file and builds frequency map
    ItemTracker() {
        loadFileData();
        createBackupFile();
    }

    /*
     * Function: loadFileData
     * Purpose: Reads input file and populates map
     */
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sebastienguerilus@MacBookAir CS210_Project3 % ./CornerGrocer
===== Corner Grocer Menu =====
1. Search for item frequency
2. Print all item frequencies
3. Print histogram
4. Exit
Choose an option (1-4): 3

Item Purchase Histogram:
Onions         **
Potatoes       ***
Pumpkins       *****

===== Corner Grocer Menu =====
1. Search for item frequency
```

OUTLINE

TIMELINE